VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

Vishwakarma Application ID

VK26-IITH-SM-0008

FIREDAMP DYNAMICS | FD-1 Borehole Sentinel

A wellhead node that watches every methane drainage borehole, shuts the leaking one before its gas drifts toward the explosive range, and counts every cubic metre that can be used.

Stage 2: Design Specification | Track: **Sustainable Mining (IIT Hyderabad)**
Sub-theme: **Worker Health, Safety & Environmental Hazard Tracking**
Team Chakra | Lead: Saikoushik Nalubola, SR University, Warangal | Members: **[add names]**

Problem Context: drained, but unverified

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Project overview, the operational pain point and who carries it

TITLE	TEAM ID	TRACK	CORE INNOVATION (1 LINE)
Firedamp Dynamics: FD-1 Borehole Sentinel	**VK26-IITH-SM-0008 Team Chakra**	**Sustainable Mining IIT Hyderabad**	Per-borehole gas purity, flow and air-ingress detection that throttles or isolates the hole on its own, and logs MRV-grade methane totals.

0
commercial coal mine methane projects running in India
Ember, 2026

2.2 Mt
methane from India's coal mines in 2025, up from 2.0 Mt
IEA, GMT 2026

~15%
of coal methane comes from underground mines making ~3.5% of coal
Ember, 2026

5–15%
methane in air is explosive; transport below 30% is poor practice
UNECE BPG, 2016

Who feels it: target beneficiary

- **Primary:** gas-drainage crews and ventilation officers at Degree-II and III underground mines, where DGMS requires methane drainage before mining (71 Degree-II and 7 Degree-III mines in 2015).
- **First pilot region:** Jharia, Raniganj and East Bokaro coalfields, home of India's only CMM pilot (Moonidih) and the Pootkee, Sawang and Chinakuri pre-feasibility studies.
- **Secondary:** coal companies drafting methane-reduction plans and net-zero commitments.

Problem statement

A drainage range is only as clean as its leakiest borehole. Air drawn in at a cracked collar or a loose standpipe dilutes the gas from every hole downstream, and pushes it toward the 5-15% band where it can explode.

Mines measure methane for safety, but periodically and without an integrated real-time system. So a failing hole stays anonymous, gas nobody can vouch for gets vented instead of used, and there are no borehole-level numbers to put in a methane plan.

FD-1 gives each borehole its own instrument and its own valve.

Scope: FD-1 handles drained gas (typically 80-95% CH4 at the seam). Ventilation air methane, at under 1%, needs thermal oxidisers and is out of scope.

Problem Context: 18 years, zero projects

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Fact check: what has been tried, and the gap that is still open

What has been done (public record)

2008 — India CMM/CBM Clearinghouse set up at CMPDI, Ranchi, with US EPA support. CMPDI is the nodal agency for CMM.

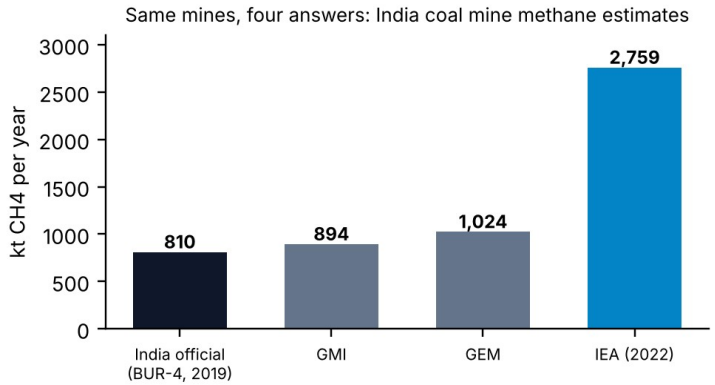
Pilot — Moonidih underground mine, Jharia: capture and use shown to be technically feasible. No commercial project followed.

Studies — Pre-feasibility studies at Pootkee, Sawang and Chinakuri collieries.

2019 — Revised rules: Degree-III mining enterprises to prepare CMM reduction plans.

2025 — Large coal companies asked to submit net-zero commitments by end-2025.

2026 — Still no national MRV framework; India's CMM estimates disagree by more than 3x.



Ember (2026) citing BUR-4, GMI, GEM and IEA. You cannot manage what you cannot measure.

Public search disclosure

- **Mine-air monitors:** protect the workings; they cannot say which borehole is leaking into the range.
- **Catalytic (pellistor) heads:** need oxygen and are made for the LEL range, not 0-100% vol drainage gas.
- **Gap:** no Indian-made unit we could find measures purity, flow and air ingress at each borehole and acts on it. Patent search is a Stage 3 deliverable.

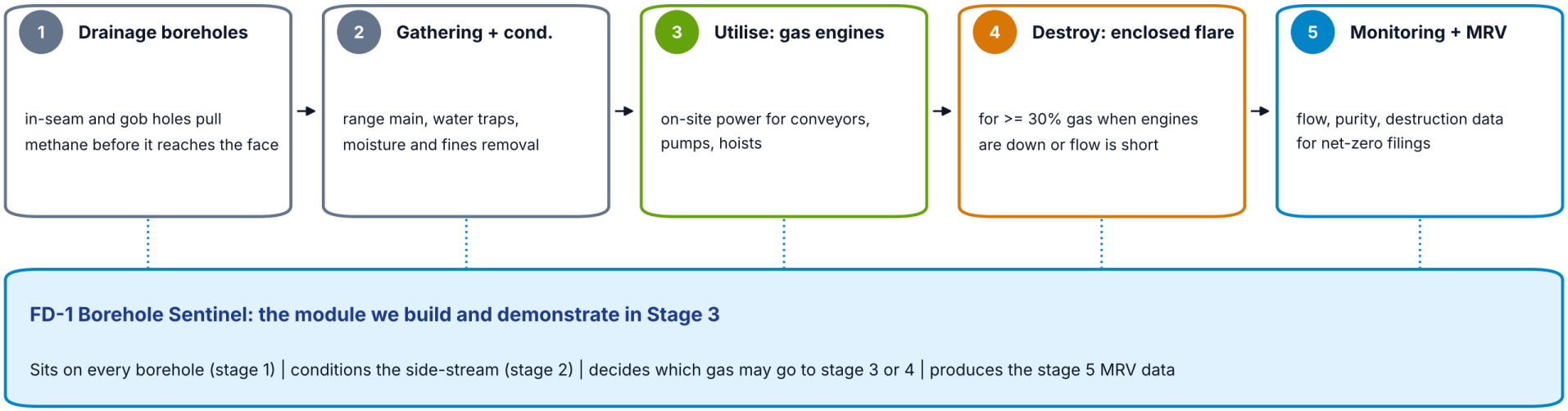
Why the gap persists: Ember (2026) names missing policy, missing incentives and missing data. FD-1 attacks the one an engineering team can fix: data at the borehole.

The Firedamp System: where FD-1 fits

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Our Stage 1 concept stays the vision. FD-1 is the module we build first, because every other stage depends on it.

Firedamp Dynamics system (Stage 1 concept) | ventilation-air flaring removed: < 1% CH4 cannot sustain a flame



Same problem, same five stages

Drainage, conditioning, utilisation, destruction and MRV, as proposed in Stage 1, sized to each mine's gas class and seam geology.

One correction

Flaring ventilation air is gone: at under 1% CH4 it cannot hold a flame and needs thermal oxidisers. The flare now burns only drained gas at 30% or more.

Why FD-1 first

Engines and flares only work if gas quality is known. UNECE notes air ingress can drag drainage gas below 30%. FD-1 is the part a student team can build and prove in four months.

Roadmap continuity: The Stage 1 plan holds: pilot cluster at 9-18 months, MRV record at 18-30, replication after 30. FD-1 is the instrument each phase needs.

User Journey: the before state

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Current pain points, mapped step by step and benchmarked

BEFORE: drained, but blind

1. **Borehole drains seam gas into the range main**
   Degree-II/III mines must pre-drain (DGMS). Seam gas starts rich, often 80-95% CH4.
2. **Air leaks in at the collar or standpipe, unseen**
   Each leaking hole dilutes the whole range. Nobody sees which hole it is.
3. **Quality checked on periodic rounds or at the plant**
   Measurements are periodic and safety-driven; no per-borehole live data.
4. **Gas of unknown quality is vented, not used**
   No commercial CMM project runs in India; capture never reaches utilisation.
5. **No borehole-level numbers for MRV or net-zero plans**
   Official estimate 810 kt CH4 (2019) vs IEA 2,759 kt (2022).

432 Nm3

CH4 vented per day by one borehole draining 0.3 m3/min of methane (illustrative)

our calc, analysis/gas_calcs.py

24.8 t

CO2e per day from that one hole at a 20-year GWP of 80

our calc; GWP framing per Ember/IEA

1,504 kWh

electricity per day left unmade, at 35% gas-engine efficiency

our calc, stated assumptions

$13.3 bn

India's bill for ~31 bcm of imported gas in FY2023-24

Ember, 2026

Safety hazard, modelled: with a collar leak growing to 8 L/min and no per-hole instrument, range purity spent 859 s of a 20-minute run below the 30% good-practice minimum and settled at 17.0% CH4, with nothing raising an alarm.
Source: analysis/leak_sim.py (plant model and assumptions in the file).

Bottom line: The gas is drilled for, piped and paid for, then thrown away because no one can prove which holes are good.

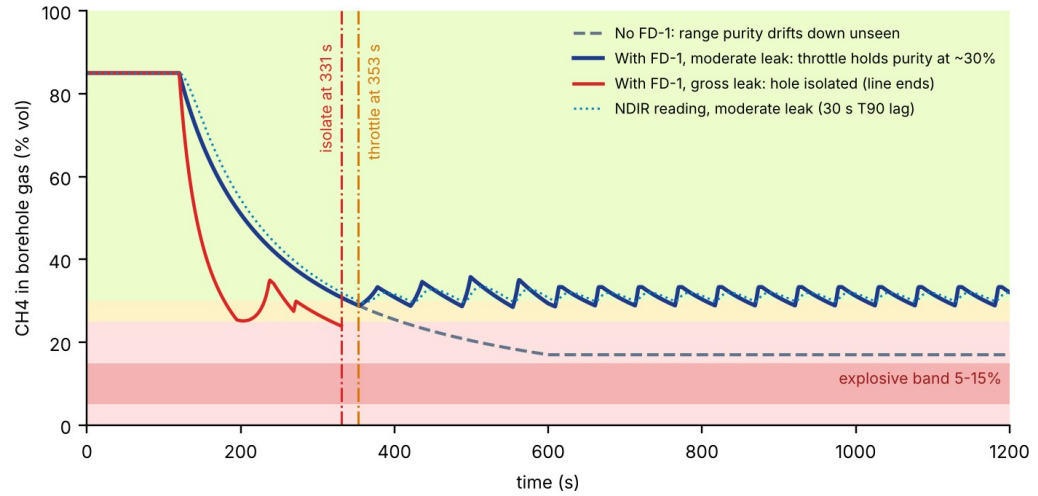
User Journey: the after state

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How FD-1 changes the same five steps, with benchmarks from our model

AFTER: every borehole has a voice

1. **FD-1 reads CH4, O2, flow, pressure at the hole**
   1 Hz sampling; two independent CH4 measures (NDIR and O2-derived).
2. **Leak caught early: O2 up, CH4 down**
   Node throttles that hole to recover purity; isolates it below 25% CH4.
3. **Only gas >= 30% CH4 (2 x UEL) moves on**
   Router sends it to a gas engine, or to an enclosed flare when there is no load.
4. **Every cubic metre is counted**
   Density-compensated flow x CH4 fraction = Nm3 CH4 per borehole per hour.
5. **MRV-ready logs for compliance filings**
   CSV per borehole: drained, utilised, destroyed, isolated minutes, tCO2e.



Same leak, three outcomes. Firmware thresholds; plant model is simple and stated in leak_sim.py.

28.5%
lowest purity with FD-1 throttling a moderate leak (vs 17.0% without)
leak_sim.py

331 s
gross leak: hole isolated automatically, before the range is diluted
leak_sim.py

14.4%
flow under-read removed by density compensation at 60% CH4
gas_calcs.py

60 s
per-borehole MRV record: drained, routed, isolated minutes
firmware design

Quantified efficiency: Detection moves from the next inspection round to one reading per second per hole; leaking holes are fixed by number, not by search.

Innovation & Patentability

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What FD-1 does that today's practice does not

Capability	Portable detector rounds	Monitoring at the extraction plant	FD-1 at every borehole
Names the leaking borehole	only if the round reaches it	no, one combined stream	yes, per hole, every second
Live drainage-gas purity	no	yes, combined	yes, per hole
Acts on its own	no	plant-level trips	throttles or isolates the hole
Checks its own sensors	no	depends on system	NDIR vs O2 cross-check
Per-borehole MRV totals	no	no	Nm3 CH4 drained and routed

Typical practice as we understand it from public sources; to be validated with mine engineers.

Patentability: draft provisional claims cover ideas 1-3 working together on one wellhead unit (docs/patent_notes.md). IP stays with the team.

Prior-art search: Google Patents and InPASS with the keywords listed in the repo; results reported in Stage 3. We claim novelty only after that search.

1 Borehole-level purity loop

Each hole holds its own gas above 30% by trimming its suction, instead of the whole range being diluted by one bad collar.

2 Two-principle self-check

Seam gas carries almost no oxygen, so CH4 is also 1 - O2/20.95. If NDIR and this estimate disagree by 5 points, the node fails closed.

3 Pump-less side-stream

The metering orifice's own pressure drop pushes gas through the sensing chamber. No sample pump to fail underground.

4 Composition-corrected MRV

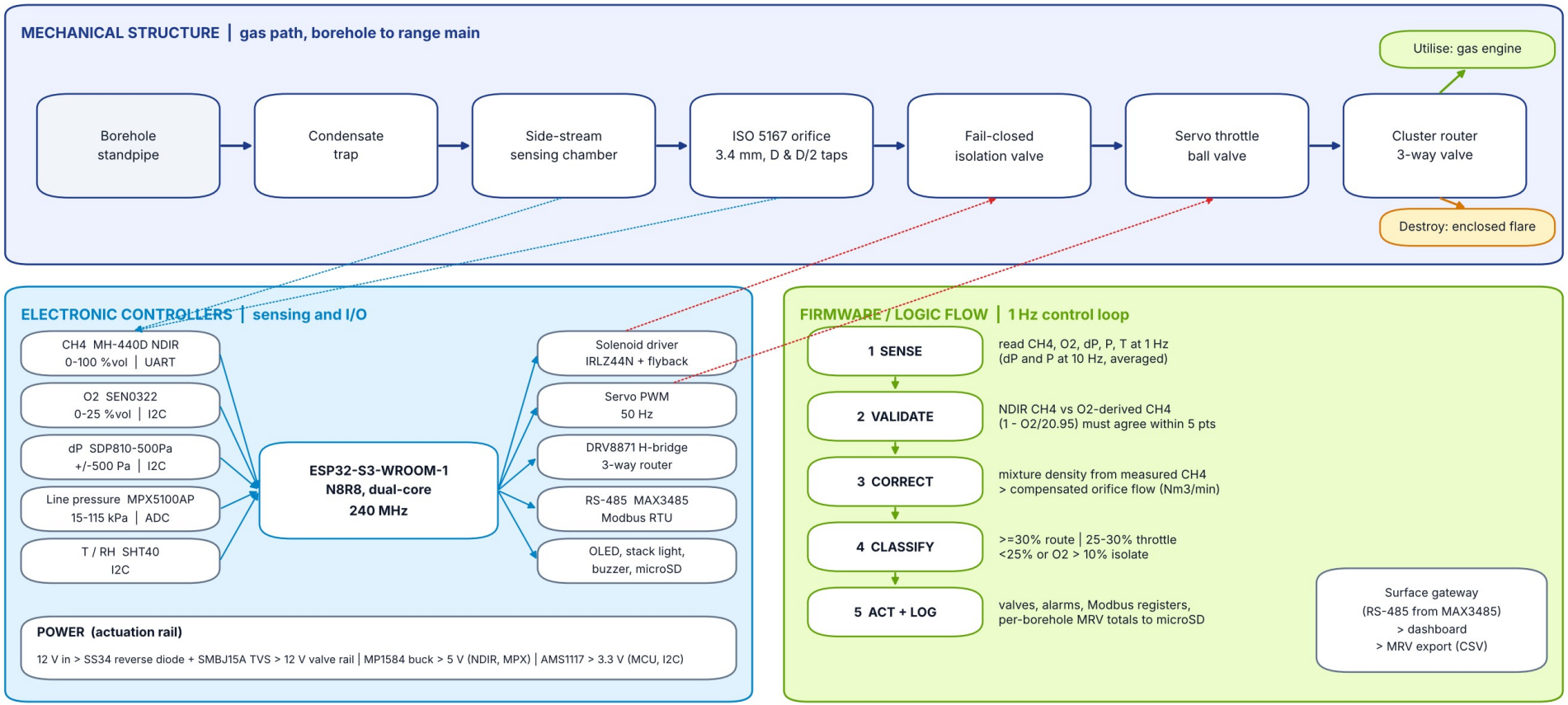
Flow is corrected with the measured mixture density, removing a 14.4% under-read at 60% CH4.

Why it matters: Engines need gas they can trust and regulators need numbers they can audit. FD-1 produces both at the point where the gas is born.

System Architecture: block diagram

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Four subsystems: mechanical gas path, actuation and power, electronic control, firmware and data



Architecture rationale: RS-485 Modbus RTU over Wi-Fi or LoRa: radio does not carry reliably through strata, and a twisted pair runs up to 1,200 m with no gateway underground.

Target Specs & Hardware Selection

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Every target maps to a named part and protocol. Bench values unless marked field.

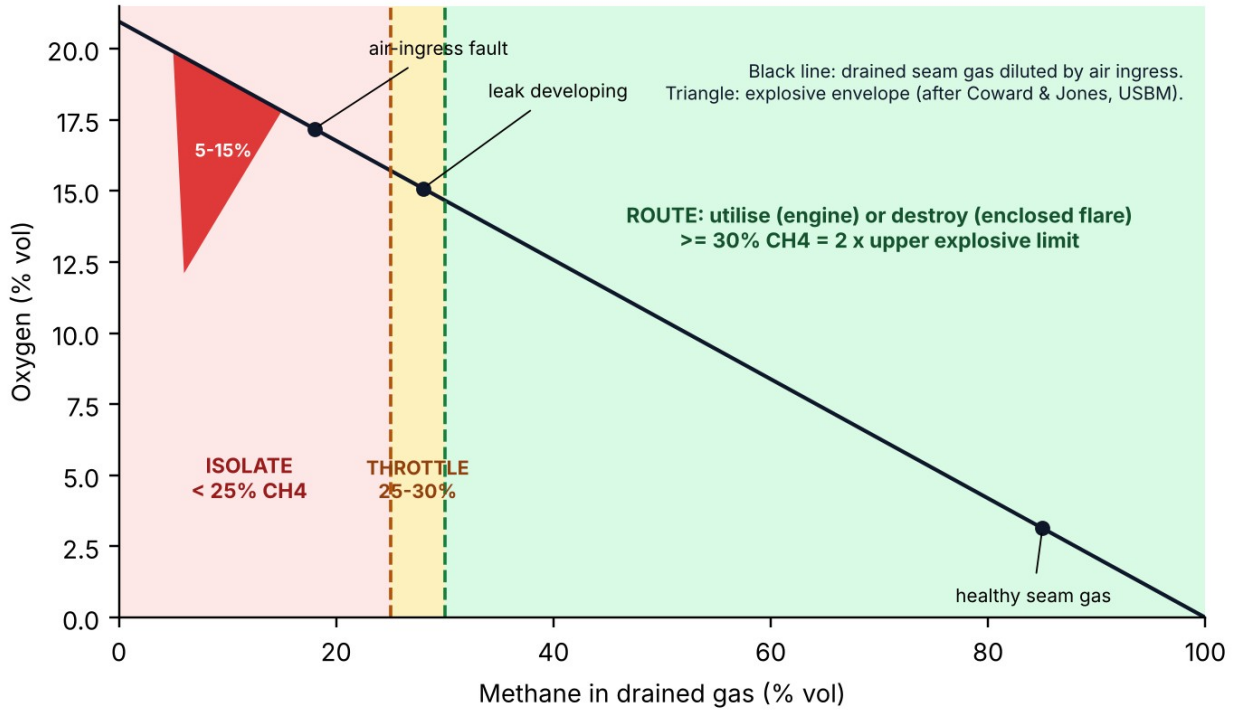
Parameter	Target spec	Selected hardware / protocol
Methane	0-100 %vol, T90 < 30 s	Winsen MH-440D NDIR, UART + 0.4-2.0 V analog
Oxygen (air-ingress signal)	0-25 %vol	DFRobot SEN0322 electrochemical, I2C 0x73
Flow	0.5-6 L/min bench, +/-5% after calibration	ISO 5167 orifice, 3.4 mm bore (beta 0.215) + Sensirion SDP810-500Pa
Line pressure	15-115 kPa absolute	NXP MPX5100AP, ADC via 10k/18k divider
Decision loop	1 Hz decisions, 10 Hz dP and pressure	ESP32-S3-WROOM-1-N8R8, dual core 240 MHz
Isolation	< 1 s to closed, closes on power loss	12 V NC direct-acting solenoid, IRLZ44N low-side
Purity bands	>= 30% route | 25-30% throttle | < 25% isolate	firmware config.h (UNECE: 2 x UEL = 30%)
Sensor cross-check	NDIR vs O2-derived CH4 within 5 points	two measuring principles, FAULT if they disagree
Comms	Modbus RTU 19.2 kbps, up to 1,200 m	MAX3485, 120 ohm termination
Power	12 V DC, <= 3 W average, <= 15 W at valve stroke	SS34 + SMBJ15A, MP1584 5 V, AMS1117 3.3 V
Data	30 days of 1-minute MRV records	microSD over SPI, CSV export via gateway
Enclosure	IP65 bench | Ex d or Ex ia, Group I (field)	ABS/PC 200×120×75 mm, PG7 glands

Selection logic: NDIR, not catalytic, because drainage gas sits far above the LEL and may be oxygen-poor. The O2 cell is the second opinion and the leak detector.

Working Process: one decision per second

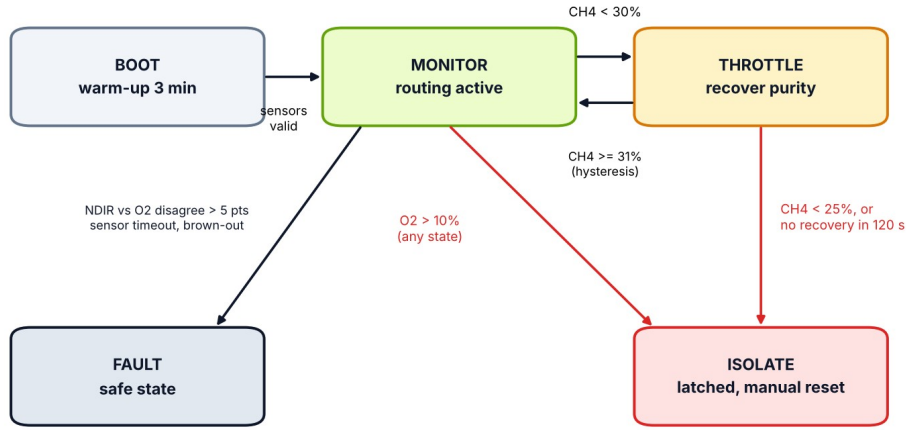
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Where the gas sits on the explosibility diagram decides what the valves do



Bands from UNECE Best Practice Guidance (2016): 30% = 2 x upper explosive limit; national minimums run 25-40%.

Explosive triangle after Coward & Jones (US Bureau of Mines), approximate vertices.



ISOLATE and FAULT both drive valves closed. Loss of power closes the NC valve by spring.

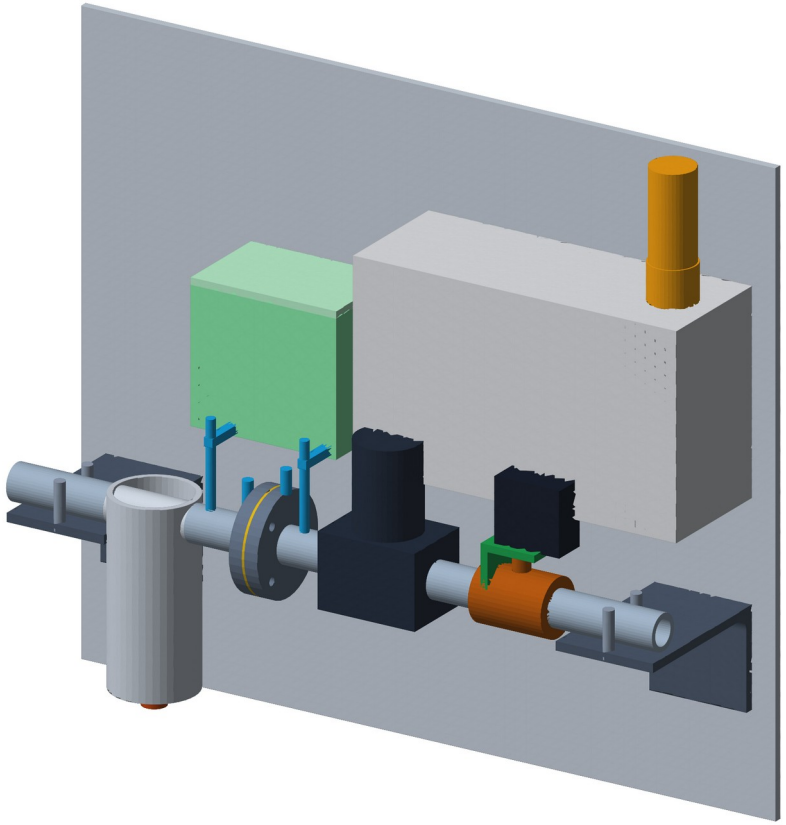
- **O2 as leak tracer:** seam gas carries almost no oxygen, so O2 rising at a hole means air is getting in there.
- **Bench trick:** with CO2 standing in for methane, purity = 1 - O2/20.95. The same equation cross-checks the NDIR in a real mine.

Fail-safe default: Any doubt ends in a closed valve: stale sensor, disagreement between sensors, brown-out or firmware hang.

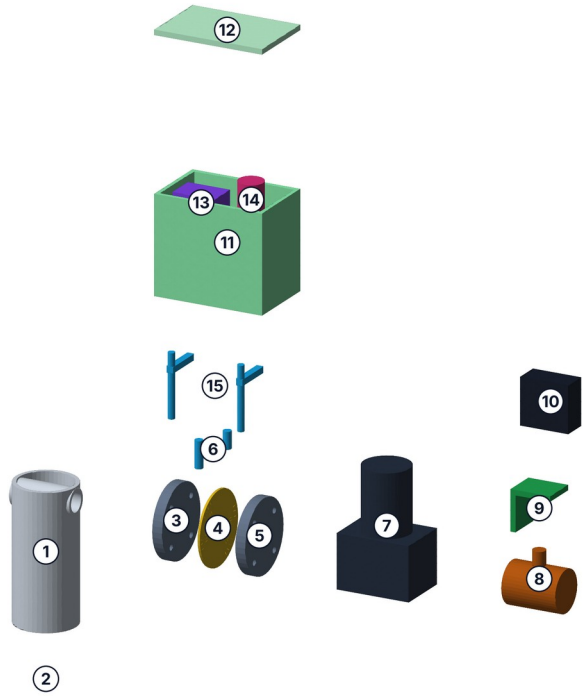
CAD Design: assembly and exploded view

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Parametric CadQuery model; 24 parts exported as STEP and STL in the repo



Isometric: bench FD-1 node on its backplate, gas flows left to right



Exploded meter run and sensing chamber

Part index

1 Condensate trap
2 Drain plug
3 Upstream flange
4 Orifice plate, SS304
5 Downstream flange
6 D and D/2 dP taps
7 NC isolation solenoid
8 Throttle ball valve
9 Servo bracket (PETG)
10 Throttle servo
11 Sensing chamber
12 Chamber lid + gasket
13 CH4 NDIR (MH-440D)
14 O2 cell (SEN0322)
15 Side-stream sample loop

Key mechanism: The orifice pressure drop drives the side-stream through the sensing chamber, so the node needs no sample pump.

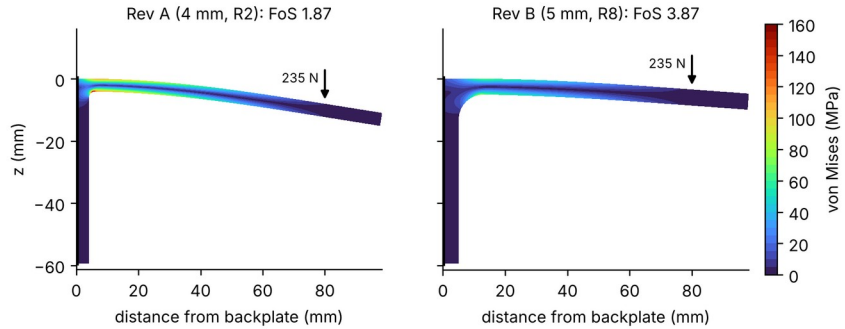
Mechanical Specs: materials and FEA

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Material choices, FEA of the pipe bracket, and hand checks on the pressure parts

Part	Bench material	Why	Field change
Orifice plate	SS304, 1.5 mm	sharp edge stays sharp in wet, acidic gas	same
Flanges, brackets	Al 6061-T6	machinable, light, quick to iterate	SS304 / mild steel*
Sensing chamber	PETG, 3 mm wall	printable, inert to CH4 and CO2	SS304 machined
Enclosure	ABS/PC, IP65	cheap, sealed, see-through window	Ex d cast enclosure

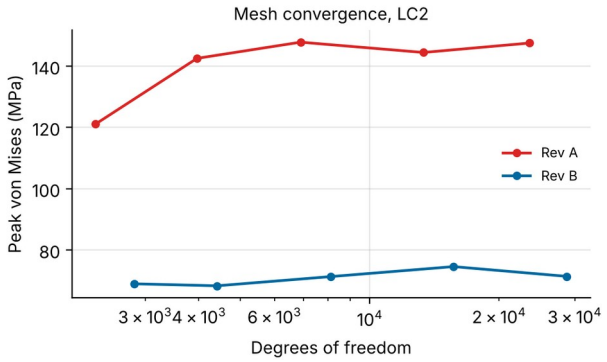
*IEC 60079-0 limits light alloys in Group I (mining) equipment because of frictional sparking. Aluminium stays on the bench.



Von Mises on one 0-160 MPa scale, deformation x5. Peak sits at the inside bend.

Validation summary

- **Bracket FEA:** plane stress, P2 triangles, 6,860 elements. 12 kg spool at 4 g on two brackets = 235 N each.
- **Rev A:** 4 mm plate, 2 mm bend radius: 147.5 MPa peak, FoS 1.87 against 276 MPa yield. Below our 2.0 target.
- **Rev B:** 5 mm plate, 8 mm bend radius: 71.4 MPa, FoS 3.87; saddle deflection 0.71 mm.
- **Chamber wall:** PETG, 90×80 mm, 3 mm, -30 kPa, fixed edges (Roark): about 7.6 MPa and 0.37 mm. FoS ~6.



Converged

Five mesh levels per revision; peak stress settles within about 5%.

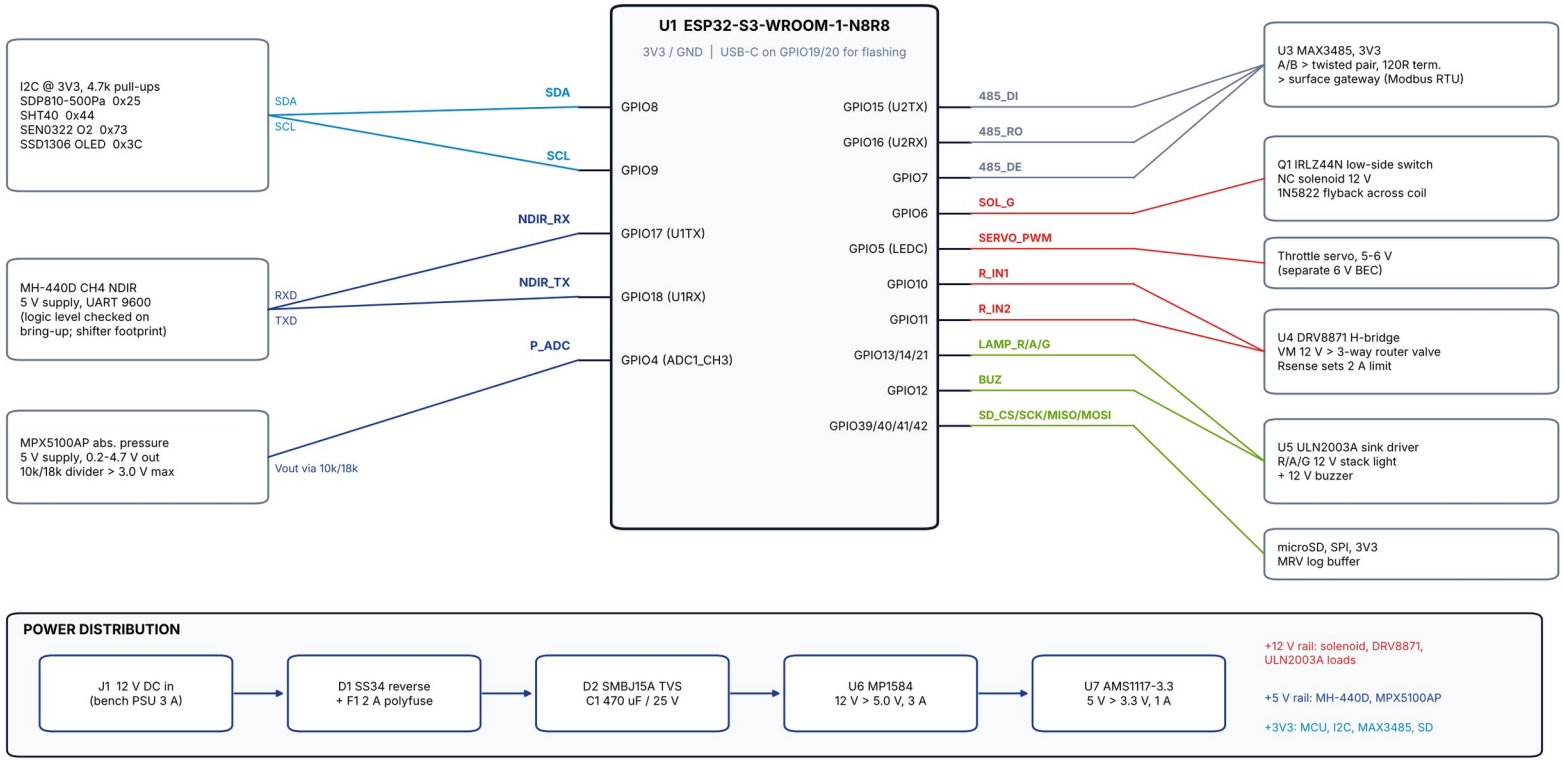
Peaks are read away from the clamped edge and load-patch corners, where idealised boundaries give non-physical spikes.

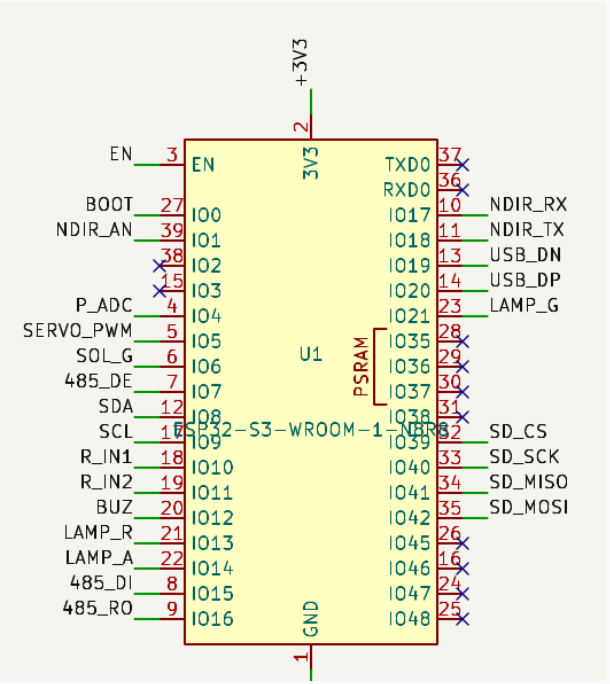
Design iteration: Bracket moved from rev A to rev B after FEA: one more millimetre and a larger bend radius doubled the margin (FoS 1.87 to 3.87).

Electronics: controller schematic

VK26-IITH-SM-0008

Pinout connections, drivers, protection and power distribution, captured in KiCad





Same design in KiCad 7 (U1 zoomed)

Native KiCad 7 file: electronics/kicad/fd1_controller.kicad_sch: 44 parts, 44 nets, netlist check passed. Native USB on GPIO19/20 for flashing; octal-PSRAM pins 35-37 left free.

Electronics: itemised prototype BOM

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Two FD-1 nodes, one router and the bench rig. Prices are pre-order estimates (INR, Oct 2026).

Component	Part / specification	Qty	Source	Lead time	Cost (INR)
MCU module	Espressif ESP32-S3-DevKitC-1 (WROOM-1-N8R8)	2	Local	2-3 days	2,400
CH4 sensor	Winsen MH-440D NDIR, 0-100 %vol	2	Imported (CN)	10-15 days	11,000
O2 sensor	DFRobot SEN0322 Gravity I2C, 0-25 %vol	2	Local distributor	3-5 days	8,400
Differential pressure	Sensirion SDP810-500Pa, I2C	2	Imported (Mouser IN)	5-7 days	7,200
Line pressure	NXP MPX5100AP, 15-115 kPa abs	2	Imported (Mouser IN)	5-7 days	2,800
Isolation valve	12 V NC direct-acting solenoid, 1/2" brass	2	Local	3-5 days	1,300
Throttle	1/2" brass ball valve + MG996R servo	2	Local	2-3 days	1,460
Enclosure	IP65 ABS/PC 200×120×75 + 4 x PG7 glands	2	Local	3-5 days	2,040
Orifice meter	SS304 plate 1.5 mm (laser cut) + 2 x Al 6061 flanges (machined)	2	Local job shop	5-7 days	2,000
PCB	2-layer FR4 1.6 mm controller board, lot of 5	1	Domestic fab	6-8 days	1,800
3-way valve	12 V motorised 3-way ball valve, 1/2"	1	Local	3-5 days	2,800
Surrogate gas	CO2 cylinder + regulator (non-flammable)	1	Local	2-3 days	3,800
Span check gas	Certified 2.5 %vol CH4 in air, small cylinder (or lab loan)	1	Domestic gas vendor	7-10 days	4,500
Other (14 lines)	T/RH, RS-485, drivers, HMI, SD, power, pump, rotameters, pipework, frame, PSU		Local	2-7 days	17,170
Total build	**full line-by-line list: electronics/bom.csv**				**68,670**

Procurement: Imported parts (MH-440D, SDP810, MPX5100AP) are ordered in week 1; everything else is local with a lead time of 7 days or less. Invoices replace estimates.

Your Maker Journey: on the bench

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Authenticity: team members at work, ID badges in frame

Insert photo
(team member wearing
institutional ID)

[Photo] Modelling the node in CAD

Insert photo
(team member wearing
institutional ID)

[Photo] Soldering the controller board

Insert photo
(team member wearing
institutional ID)

[Photo] Assembling the bench rig

Insert joint lab review photo with faculty mentor (if attached)
Caption: name, department, date

Stage 1 to Stage 2 engineering pivots

- **Plant stays, node first:** the Stage 1 five-stage system is still the vision. Vishwakarma rewards a working prototype, so we build its weakest link first: gas quality at the borehole.
- **VAM flaring dropped:** air at under 1% CH4 cannot hold a flame; it needs thermal oxidisers. FD-1 handles drained gas only.
- **Pilot moved:** SCCL tender documents describe the seams at the mines we checked as Degree-I. First field target: Degree-II/III mines in Jharia and Raniganj.
- **Safe bench:** CO2 surrogate plus O2-derived purity, so no flammable gas sits on a college bench.
- **Bracket:** rev A to rev B after FEA, FoS 1.87 to 3.87.

Stage 2 design milestones (Oct 2026)

24-part parametric CAD, STEP + STL exported
Bracket FEA with 5-level mesh convergence
Closed-loop leak model on firmware thresholds
Pin map, BOM and firmware baseline in repo

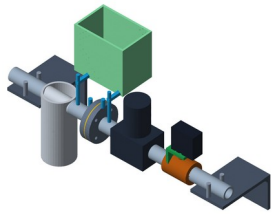
Institutional sign-off: [Faculty guide name, department, SR University: add after review]

Your Maker Journey: design evidence

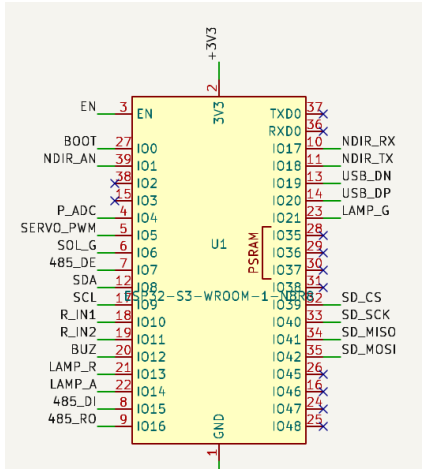
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Original design assets and the raw repository

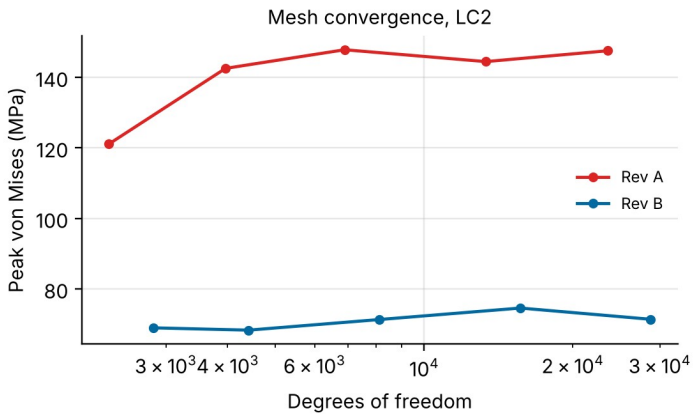
```python
P = dict(
    pipe_od=21.3,        # 1/2" NB pipe (DN15)
    pipe_id=15.8,
    orifice_bore=3.4,    # ~6 L/min of air at ~200 Pa (analysis/gas_calcs.py)
    flange_d=60.0,
    flange_t=8.0,
    plate_t=1.5,         # SS304 orifice plate
    bolt_pcd=44.0,
    bolt_d=5.5,          # M5 clearance
    trap_od=50.0,
    trap_id=44.0,
    trap_h=110.0,
    enc_w=200.0, enc_h=120.0, enc_d=75.0,   # IP65 ABS/PC electronics enclosure
    chamber_w=90.0, chamber_h=80.0, chamber_d=60.0,
    backplate_t=4.0,
    bracket_t=5.0, bracket_w=60.0,   # rev B after FEA (analysis/bracket_fea.py)
    bracket_r=8.0,                   # inside bend radius, rev B
)
```



CAD: parameter block drives all 24 parts (fd1_assembly.py)



EDA: KiCad 7 schematic, netlist check 0 errors



FEA convergence; mesh 3,754 nodes / 6,860 elements (rev B)

Raw repository linkage

Repository: github.com/saikoushiknalubola/firedamp-fd1

Native files: cad/step/*.step (24 parts + assembly), cad/stl, cad/src (parametric model), electronics/kicad (native .kicad_sch, netlist, checker), electronics/ (pin map, BOM), firmware/fd1_node (ESP32 code), analysis/ (FEA, orifice, leak model).

Audit trail: every figure in this deck is generated by a script in the repo, so commit timestamps trace the work through Stage 2. No embedded videos.

Next in EDA: Footprint assignment and 2-layer PCB layout in KiCad, then DRC; imported sensors ordered in week 1 of fabrication.

Safety & Hazard Mitigation

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Physical fail-safes, design FMEA and the standards that shape the field version

Failure mode	Potential hazard	Embedded fail-safe / mitigation
Air ingress at collar or standpipe	range gas drifts toward 5-15%	throttle that hole; isolate below 25% CH4 or above 10% O2 (latched)
NDIR drift or failure	wrong routing decision	O2-derived cross-check; > 5 points apart = FAULT, valve closed; span check with 2.5% CH4 gas
Loss of 12 V / brown-out	valves left in unknown state	NC solenoid spring-closes; MCU brown-out reset restarts in BOOT (closed)
Firmware hang	valve stuck open	hardware watchdog resets to closed state within 3 s
Water slug in the line	flooded sensors, false readings	condensate trap with drain; hydrophobic PTFE filter on sample inlet
Flame front in pipeline (field)	propagation along the range	in-line flame arrestor at node outlet (UNECE protection measure)
Surface link lost	no visibility above ground	node keeps acting locally, logs to microSD, amber lamp

Standards compliance

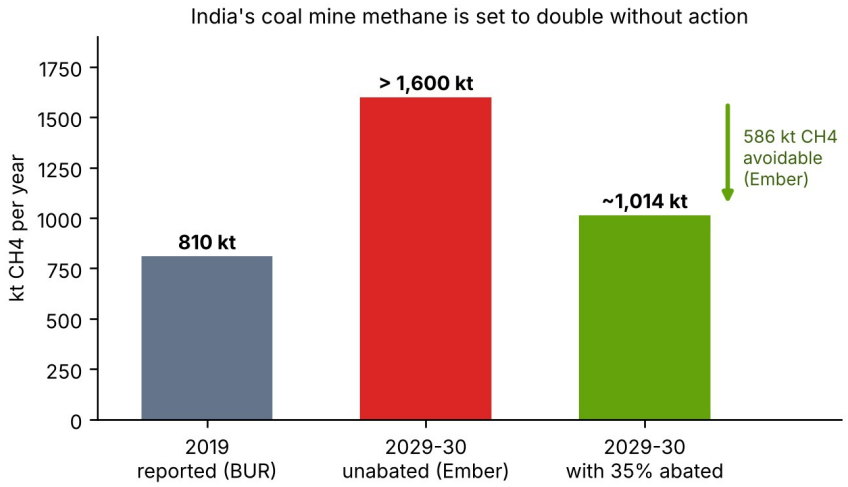
- **IS/IEC 60079-0, -1, -11:** Group I equipment: flameproof (Ex d) and intrinsic safety (Ex ia) path
- **IEC 60079-29-1:** performance of flammable gas detectors
- **ISO 5167-2:** orifice plates, D and D/2 tappings
- **IEC 60529:** IP65 enclosure, PG7 glands
- **Coal Mines Regulations 2017:** DGMS approval before any equipment goes underground
- **UNECE BPG (2016):** 30% minimum for transport and use

Physical fail-safes: Emergency stop on the bench rig cuts 12 V; the bench prototype is not certified and will not be taken underground.

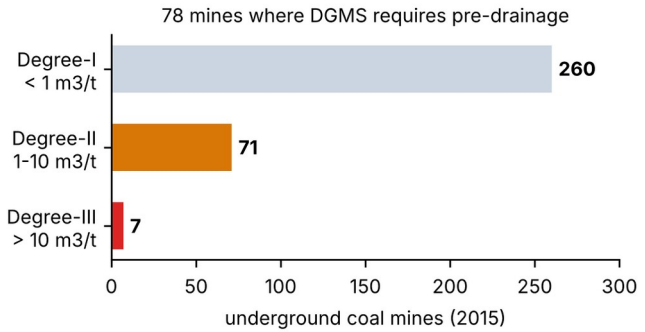
Impact & Commercial Viability

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Size of the problem, value of one borehole, and what a node costs



Ember (2025, 2026); 586 kt avoidable by 2029.



Mine counts for 2015 (MoLE 2017, via Sadavarte et al.).

$612/t — value of captured CH4 at Ember's import-offset rate ($980 mn for ~1,600 kt)

$190/day — gas value of one borehole draining 0.3 m3/min CH4 (illustrative)

Rs 21.7k — parts cost of one bench FD-1 node (BOM)

< 2 days — for one borehole's gas value to cover that, at Rs 85-100 per $

Impact by pillar

- **Workers:** a leaking hole is named and shut within the sensor response time, not at the next round.
- **Climate:** 24.8 tCO2e a day per borehole kept out of the air when its gas is used or flared (20-yr GWP).
- **Energy:** about 1,504 kWh a day per borehole at 35% engine efficiency, offsetting diesel and grid draw.

Ease of use and deployment

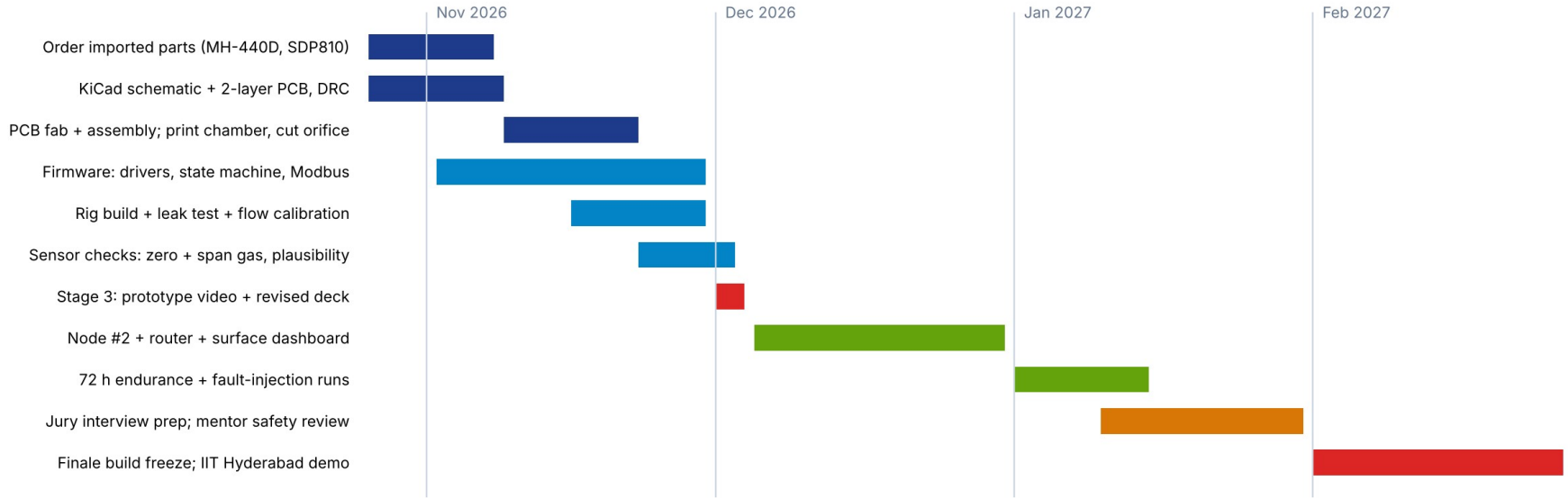
- **Install:** one spool on the standpipe, one cable to the gateway; no change to drilling.
- **Operate:** traffic-light lamp at the hole; alarms and totals on the surface screen.
- **SDGs:** 8 Decent Work, 12 Responsible Production, 13 Climate Action.

Field cost honesty: An Ex-certified field unit will cost much more than the bench node. Even at 10 times the bench cost, one borehole's gas covers it in under two weeks.

Roadmap: 4-month fabrication plan

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From design freeze to the IIT Hyderabad finale



Fabrication (Nov 2026)

- **Week 1-2:** order imported sensors; KiCad capture and 2-layer PCB with DRC.
- **Week 3-4:** board assembly, printed chamber, laser-cut orifice plate; rig plumbing, leak test, flow calibration.

Stage 3 commitment (Dec 2026)

One complete FD-1 node on the rig, running leak, throttle and isolate end to end with CO2 surrogate; flow within +/-5% of the rotameter; NDIR span-checked; recorded on video with the revised deck.

After Stage 3: Second node, router and surface dashboard in December; 72-hour endurance and fault-injection runs in January; build freeze before the February finale.

Technical Challenges & Risks

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Engineering bottlenecks found in simulation and design review

Sensor lag

Observed risk: NDIR T90 is 30 s. In the gross-leak run the hole was isolated at 331 s, but true purity touched 24.0% first, just under the 25% line.

Severity: high

Throttle limit cycle

Observed risk: on-off throttling plus sensor lag makes purity saw-tooth between about 29% and 35% (leak_sim.py). It holds the band but wears the valve.

Severity: medium

Wet, dirty gas

Observed risk: drainage lines carry water and coal fines. Both can blind an optical sensor and block a 4 mm sample loop.

Severity: medium

Going underground

Observed risk: the bench build is IP65, not Ex certified. Field use needs Ex d or Ex ia design, steel hardware and DGMS approval.

Severity: high (field)

Risk	Likelihood	Impact	Early warning we can measure
Sensor lag lets a fast leak dip below 25%	Medium	High	dP jump with falling line vacuum ahead of NDIR change
Throttle valve wear from cycling	High	Medium	valve travel per hour logged by firmware
Water or fines reach the NDIR	Medium	Medium	NDIR vs O2 disagreement rising; condensate pot level
Field approval takes longer than the programme	High	High (field only)	none on bench; tracked as a separate path

Site data risk: No mine-level gas data yet. Bench flow ranges are set from published pre-feasibility studies until a mine shares numbers.

Potential Solutions & Risk Mitigation

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Backup technical solutions for each bottleneck

Fast leading indicator

Mitigation: suction pressure and dP sample at 10 Hz. A sudden flow rise with falling line vacuum is the leak signature, so the throttle moves before the NDIR catches up.

PI throttle loop

Mitigation: replace on-off steps with PI control on purity error, with anti-windup; tune on the rig and compare valve travel per hour.

Protect the optics

Mitigation: condensate trap, low-point drain, hydrophobic PTFE membrane at the sample inlet, and a timed back-purge.

Certification path

Mitigation: low-power electronics laid out for an Ex ia barrier, Ex d enclosure, steel spool; partner with a DGMS-approved manufacturer.

Mitigation	Bench test (Stage 3)	Pass criterion (target)
Fast leading indicator	step air-bleed valve open; repeat 10 times	hole isolated before true purity < 25% in 10 of 10 runs
PI throttle loop	1 h moderate-leak run, on-off vs PI	purity held at 30% or above with at least 50% less valve travel
Protect the optics	inject a 50 mL water slug upstream	no liquid in the chamber; reading back within 2 min
Certification path	design review against IEC 60079-11 with mentor	written gap list for the Ex ia version

Sensor supply fallback: a second-source 0-100 %vol NDIR module on the same UART/analog footprint. | Data fallback: Pootkee, Sawang and Chinakuri pre-feasibility reports.

Verification: Each mitigation has a bench test in the Stage 3 plan: fault injection for lag, valve-travel logging for the PI loop, water-slug test for the sample loop.

References & Sources

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7. Sadavarte et al., gridded CMM inventory for India and Australia, arXiv:2107.10317 (citing MoLE 2017 mine counts).
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11. Maker Bhavan Foundation, Vishwakarma Awards 2026-27: themes and judging criteria. vishwakarma-awards.org
12. Roark's Formulas for Stress and Strain, rectangular plates with fixed edges (chamber wall check).

Image sources: every figure is generated from source files in the repository (CadQuery model, Python analysis and diagram scripts). No stock, third-party or image-generator pictures are used. Logos: Vishwakarma Awards / Maker Bhavan Foundation template.

Analysis files: analysis/gas_calcs.py, analysis/bracket_fea.py, analysis/leak_sim.py, electronics/bom.py: rerun any script to reproduce a number.